Xiaoyan Wu

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Academic Positions

University of Zurich Psychiatric University Hospital Zurich — Dec 2024–Present
ETH Zurich Neuroscience Center Zurich — Zurich, Switzerland
Postdoctoral Fellow in Computational Psychiatry. Supervisor: Philipp Homan

Education Background

Beijing Normal University, School of Psychology — Oct 2024
PhD in Cognitive Neuroscience. Supervisor: Chao Liu and Hang Zhang — Beijing, China
University of Zurich, Zurich Center for Neuroeconomics — Jan 2024
PhD in Neuroeconomics (2 years Joint Training). Supervisor: Christian C. Ruff — Zurich, Switzerland
Shenzhen University, Department of Psychology — Jun 2020
MSc in Psychology. Supervisor: Yuejia Luo — Shenzhen, China
Zhuhai College of Jilin University, Business School — Jul 2017
BS in Economics and Business Administration — Zhuhai, China

Research Training

- MRI User Certification, **University Hospital Zurich**, Zurich, Switzerland — Jun 2023
- Neuroeconomics Summer School, **University of Pennsylvania**, PA, USA — Jul 2023
- Computational Psychiatry Course, **University of Zurich**, Zurich, Switzerland — Sep 2022

Selected Awards

- Best Poster Presentation Award — 2026
  20th Alpine Brain Imaging Meeting Conference, Champéry, Switzerland
- Best PhD Dissertation Award (Top 1%) — 2025
  Beijing Normal University, Beijing, China.
- Best Dissertation Award Nominee — 2025
  Annual Society for Neuroeconomics Meeting. Cambridge/Boston, Massachusetts, USA
- Paper of the Year Award Nominee — 2025
  Annual Society for Neuroeconomics Meeting. Cambridge/Boston, Massachusetts, USA
- S4SN Travel Award — 2024
  Travailing Award of Society for Social Neuroscience annual meeting, Tsukuba, Japan.
- Outstanding Graduates (Top 1%) — 2020
  Outstanding graduate of Shenzhen University.
- The First Prize Scholarship (Top 5%) — 2019
  An academic award that recognizes outstanding achievements during my master's studies.
- National scholarship (Top 1%) — 2018
  The highest academic award given by the Chinese government to exceptional undergraduate and postgraduate students.

Publications † Co-first authors. * Co-last authors.

Peer-Reviewed Publications

- **Wu, X.**†, Fu, H.†, Aydogan, G., … & Liu, C. (2026). The self-interest of adolescents overrules cooperation in social dilemmas. ***eLife***, 14, RP106840. DOI (Press release, *eLife*)
- **Wu, X.**†, Ren, X.†, Liu, C., & Zhang, H. (2024). The motive cocktail in altruistic behaviors. ***Nature Computational Science***, 4(9), 659–676. DOI (Journal cover, *Nature Computational Science*)
- Wang, H., **Wu, X.**, … & Qin, S. (2024). Acute stress during witnessing injustice shifts third-party interventions from punishing the perpetrator to helping the victim. ***PLOS Biology***, 22(5), e3002195. DOI (Primer article, *PLOS Biology*)
- **Wu, X.**, Zhu, R., & Liu, C. (2024). A cognitive computational mechanism for mutual cooperation: The roles of positive expectation and social reward. ***Acta Psychologica Sinica***, 56(8), 1001–1015. DOI
- Jin, K., Wu, J., Zhang, R., Zhang, S., **Wu, X.**, Wu, T., Gu, R., & Liu, C. (2024). Observing heroic behavior and its gender difference in immersive virtual environments. ***Proceedings of the National Academy of Sciences of the United States of America***, 121(17), e2314590121. DOI
- **Wu, X.**, Zhu, R., Gong, X., Luo, Y., & Liu, C. (2023). Social incentives foster cooperation through guilt aversion: An effect that diminishes with primary psychopathic traits. ***PsyCh Journal***, 12(3), 417–428. DOI
- **Wu, X.**†, Feng, C.†, He, Z., Gong, X., Luo, Y. J., & Luo, Y. (2019). Gender-specific effects of vasopressin on human social communication: An ERP study. ***Hormones and Behavior***, 111, 1–9. DOI
- **Wu, X.**, Xu, P., Luo, Y. J., & Feng, C. (2018). Differential effects of intranasal vasopressin on the processing of adult and infant cues: An ERP study. ***Frontiers in Human Neuroscience***, **12**, Article 329. DOI
- **Wu, X.**, Feng, C., Xu, J., He, Z., Luo, Y., & Luo, Y. (2019). The effects of vasopressin on human social behaviors. ***Advances in Psychological Science***, 27(8), 1349–1360. DOI
- Duan, L., ... **Wu, X.**, ... Xu, P. (2018). Wavelet-based method for removing global physiological noise in functional near-infrared spectroscopy. ***Biomedical optics express***, 9(8), 3805-3820. DOI

Manuscripts Under Review

- **Wu, X.**, Wu, P., Hinzen, W., Sommer, I. E., & Homan, P. (2026). Teaching others reveals hidden structure in conceptual cognitive maps. ***bioRxiv***. DOI
- **Wu, X.**, Rabe, F., … Homan, P. (2026). Disorganization impairs cognitive maps built from visual inputs. Accepted by ***Schizophrenia Bulletin***. DOI
- Omlor, W., … **Wu, X.**, … & Homan, P. (2025). Goal-directed control in schizophrenia: Loss-biased engagement of the anterior cingulate relates to negative-symptom outcomes. Accepted by ***Schizophrenia Bulletin***. DOI
- Fu, H., … & **Wu, X.***, Liu, C.* (2025). Reciprocity reward sustains human cooperation with diverse agents under framed social intent-induced expectation. *Under review.*
- **Wu, X.**, Ren, X., Dreher, J., Liu, C. (2025). The motive cocktail in children's altruistic behaviors. *Biorxiv.* DOI

Research Funding

- University of Zurich Postdoc Grant, Zurich, Switzerland. Sep 2026–Aug 2027
  PI, 100,000 CHF

Supervision of Students

- Master's Thesis, **University of Zurich** Aug 2025–Present
  Language features from memory-based speech predict schizotypy.

- Bachelor's Thesis, **Beijing Normal University** Jun 2021–Jul 2022
  Computational mechanisms of favoritism in reciprocity (**Outstanding Dissertation**).
- Bachelor's Thesis, **Beijing Normal University** Jun 2023–Jul 2024
  Dynamic learning in human-AI cooperative interaction (**Outstanding Dissertation**).

Teaching Activities

- Invited Lecturer, **National University of Singapore** Jul 2024
  Methods in Psychological Research and Neuroscience (Summer School).
- Invited Lecturer, **University of Zurich** Aug 10, 2023
  Methods in Cognitive Neuroscience (Summer School). Lecture on functional magnetic resonance imaging in cognitive neuroscience.
- Teaching Assistant, **Beijing Normal University** Sep 2020–Jan 2021
  Social and Affective Neuroscience (Graduate course). Teaching assistant to Prof. Chao Liu.

Selected Conference Presentations

- **Wu, X.**, Rabe, F., ..., & Homan, P. Disorganization impairs cognitive maps built from visual inputs. *Schizophrenia International Research Society*, Florence, Italy, 2026.
- **Wu, X.**, Rabe, F., ..., & Homan, P. Disorganization impairs cognitive maps built from visual inputs. *Alpine Brain Imaging Meeting 2026, 20th Anniversary Edition*, Zürich, Switzerland, 2026.
- **Wu, X.**, Rabe, F., ..., & Homan, P. Cognitive disorganization impairs the ability to construct and use cognitive maps formed from visual perceptual inputs. *8th Burghölzli Psychiatry Meeting*, Zürich, Switzerland, 2025.
- **Wu, X.**, Ren, X., Liu, C., & Zhang, H. The neural signature of the motive cocktail in altruistic behavior. *21st Annual Society for Neuroeconomics Meeting*, Cascais, Portugal, 2024.
- **Wu, X.**, Bürgi, N., & Ruff, C. Cognitive components of trust. *Society for Social Neuroscience Annual Meeting*, Tsukuba, Japan, 2024.
- **Wu, X.**, Ren, X., Liu, C., & Zhang, H. The motive cocktail in third-party punishment and helping. *21st Annual Society for Neuroeconomics Meeting*, Vancouver, Canada, 2023.
- **Wu, X.**, Bürgi, N., & Ruff, C. Cognitive components of trust. *21st Annual Society for Neuroeconomics Meeting*, Vancouver, Canada, 2023.

Personal skills

- Programming: MATLAB, Python, and R. *View code samples.*
- EEG, MEG, fMRI, and eye-tracking.
- Excellent communication skills in spoken and written English.
- Native speaker of Mandarin, Cantonese, and Hokkien; A2 level in German.

Invited Talks

- Neurocomputational Components of Trust
  Invited by Prof. Jean-Claude Dreher at *Centre national de la recherche scientifique, Lyon* Jul. 2026
- From perception to structure: cognitive disorganization is linked to weaker relational knowledge construction
  Invited by Prof. Raymond Dolan at *University College London, London* Jun. 2026
- Disorganization impairs cognitive maps built from visual inputs
  Invited by Prof. Seongmin Park at *Virginia Tech, Blacksburg* Feb. 2026
- Schizophrenia spectrum disorder and cognitive maps: new findings and future directions
  Invited by Prof. Todd Hare at *University of Zurich* Dec 2025
- Quantifying language deficits in schizophrenia via neural abstract space mapping
  Invited by Prof. Wolfram Hinzen at *Universitat Pompeu Fabra* Jun. 2025

- The socioeconomic motives for third-party punishment and help
  Invited by Prof. Kazuhisa Takemura at *Waseda University* Mar 2024
- The "Motive Cocktail" in altruistic behaviors
  Invited by Prof. Yohsuke Ohtsubo at *University of Tokyo* Mar 2024
- Neurocomputational components of trust
  Invited by prof. Brian Knutson at *Stanford University* Jan 2024
- The "Motive Cocktail" in altruistic behaviors
  Invited by prof. Philippe Tobler at *University of Zurich* Dec 2022
- The motivations underlying human third-party punishment and help
  Invited by prof. Hang Zhang at *Peking University* Aug 2022

Professional Service

- Member, Education Committee, Schizophrenia International Research Society 2026–2029
- Member, Global Representation Committee, Schizophrenia International Research Society 2026–2029

  Reviewer for *eLife*, *Schizophrenia Bulletin*, *Evolutionary Psychological Science*, *Advances in Psychological Science, Collabra: Psychology*, *Frontiers in Psychology*, *Frontiers in Behavioral Economics*, and *Frontiers in Genetics*.